

Biomedical PCB for CrashSavers Trauma Training Simulator

Sensor-Integrated Hardware for Medical Simulation and Training

Project Overview

This project focused on the design and development of a custom printed circuit board for a biomedical training simulator developed for CrashSavers Trauma (https://www.appropedia.org/CrashSavers_Trauma), in collaboration with Digital Strings (<https://digistrings.com/>) in Guatemala. The system was intended to support hands-on medical training by integrating physical interaction with electronic sensing and digital feedback.

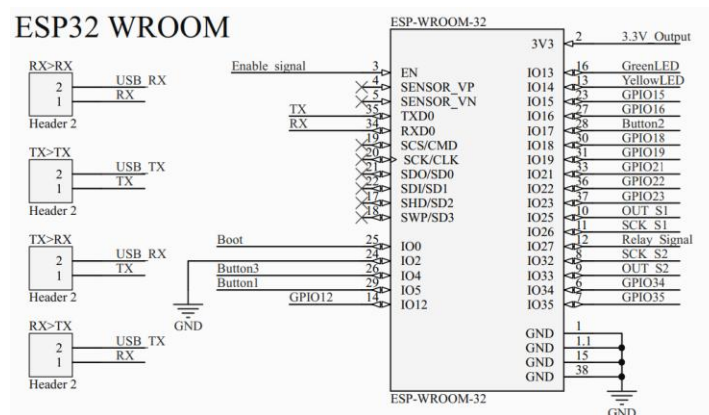


Figure 1. PCB simplified schematic block diagram.

The PCB was designed as part of a training platform that combines a physical simulator with digital educational tools. The hardware provided a way to capture user interaction with the simulator and send sensor-based information to the system, enabling a more interactive and feedback-driven training experience.

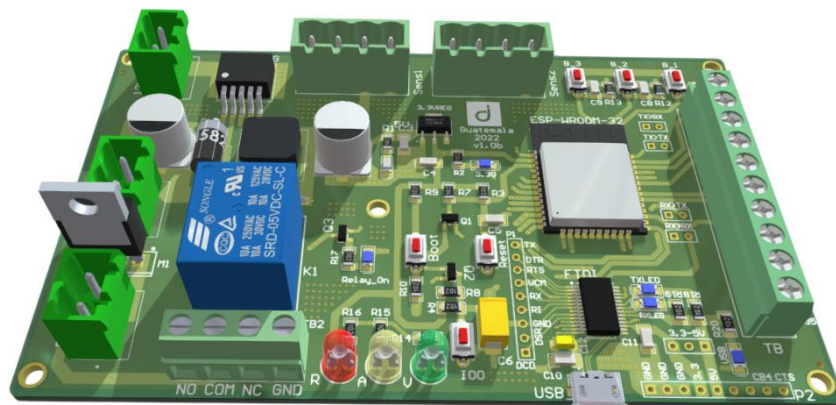


Figure 2. PCB rendered board design

Motivation

Medical simulation is an important tool for training healthcare workers, first responders, and students in a safe and controlled environment. In low-resource settings, accessible training tools can be especially valuable because they allow learners to practice critical skills without relying only on expensive commercial simulators.

The goal of this project was to contribute to a low-cost, practical, and manufacturable hardware system that could be integrated into a biomedical training simulator. The electronic design needed to be compact, reliable, easy to assemble, and compatible with the physical model and software environment used by the training platform.

System Concept

The system was based on a custom PCB designed to interface sensors, electronic components, and a microcontroller-based control system. The board intended to collect physical interaction data from the simulator and support communication with the broader training platform.

My Role

My contributions to this project included:

- Designed the custom PCB for the biomedical training simulator.
- Selected electronic components according to the project requirements.
- Prepared the schematic and PCB layout for manufacturing.
- Coordinated the PCB manufacturing process.
- Verified the manufactured board and checked basic electrical functionality.
- Programmed the control logic required for sensor acquisition and system operation.
- Supported debugging, validation, and iterative improvement of the electronics.

Methodology

The development process began by defining the functional requirements of the simulator hardware. Based on these requirements, the electronic architecture was designed, including sensor inputs, control electronics, power connections, and communication interfaces.

Components were selected according to availability, size, reliability, and compatibility with the simulator design.

After the schematic and PCB layout were completed, the board was sent for manufacturing. Once received, the PCB was inspected, assembled, and electrically verified. Firmware was then developed to read sensor signals, process the acquired data, and support the interaction between the physical simulator and the digital training platform. The final stage involved testing the PCB inside the simulator and debugging the system to improve reliability.

Key Technical Areas

This project demonstrates experience in:

- Schematic and PCB design
- Electronic components selection
- Firmware development
- Hardware validation
- System-level testing
- Prototype development

Results and Demonstrations

The project resulted in a custom PCB designed specifically for integration into a biomedical training simulator. The board supported sensor-based interaction and contributed to the development of a more interactive training tool.

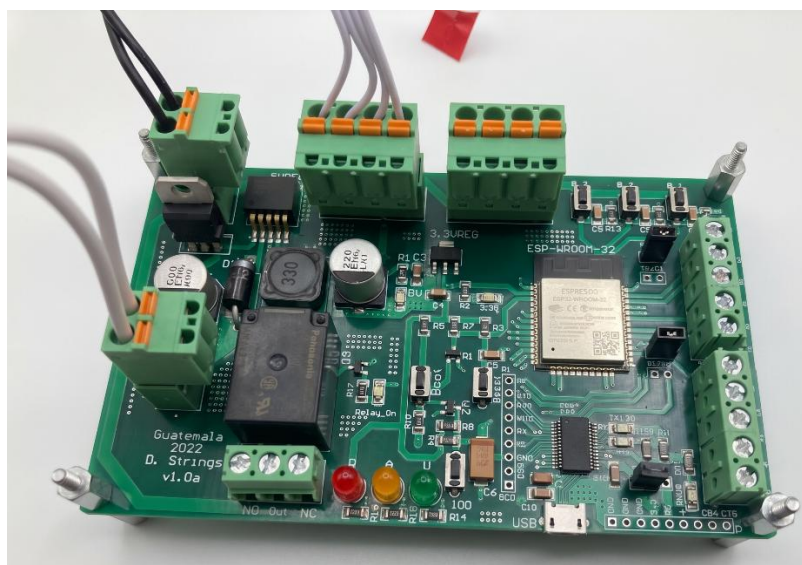


Figure 3. Manufactured PCB before integration.

Industrial and Social Relevance

This project is relevant to biomedical engineering, medical simulation, and accessible healthcare training technologies. By combining custom electronics with a physical simulator and digital educational tools, the project supports the development of practical training systems for real-world clinical and prehospital education contexts.

The project also demonstrates how engineering design can contribute to healthcare education by creating hardware that is functional, manufacturable, and adapted to the needs of a specific training environment.

Confidentiality Note

Some technical details, design files, firmware, and implementation details are omitted or generalized due to confidentiality and intellectual property considerations. The project is described here as a biomedical PCB and sensor-integration project for a medical training simulator.

Project Status

This project was developed for Digital Strings in Guatemala as part of the CrashSavers Trauma training platform. The project contributed to the development of an interactive biomedical training system that integrates physical simulation with digital educational tools.

Useful Links

Digital Strings:

- <https://digistrings.com/>

Crash Savers:

- https://www.appropedia.org/CrashSavers_Trauma
- <https://www.surghub.org/course/crash-savers-trauma>
- <https://play.google.com/store/apps/details?id=com.CrashesaversVR.CrashSavers&pli=1>